

Preliminary Study Report

Preliminary Study Synthesis

Matrix: matrix_2016

Focus: SMAP Assimilation Impact (2016)

Experiments Analyzed:

- OPL_noirr_2016
- DA_nocdf_noirr_2016
- DA_cdf_noirr_2016

Figure 1 - SMAP Assimilation Diagnostics (Jan-Jun 2016)

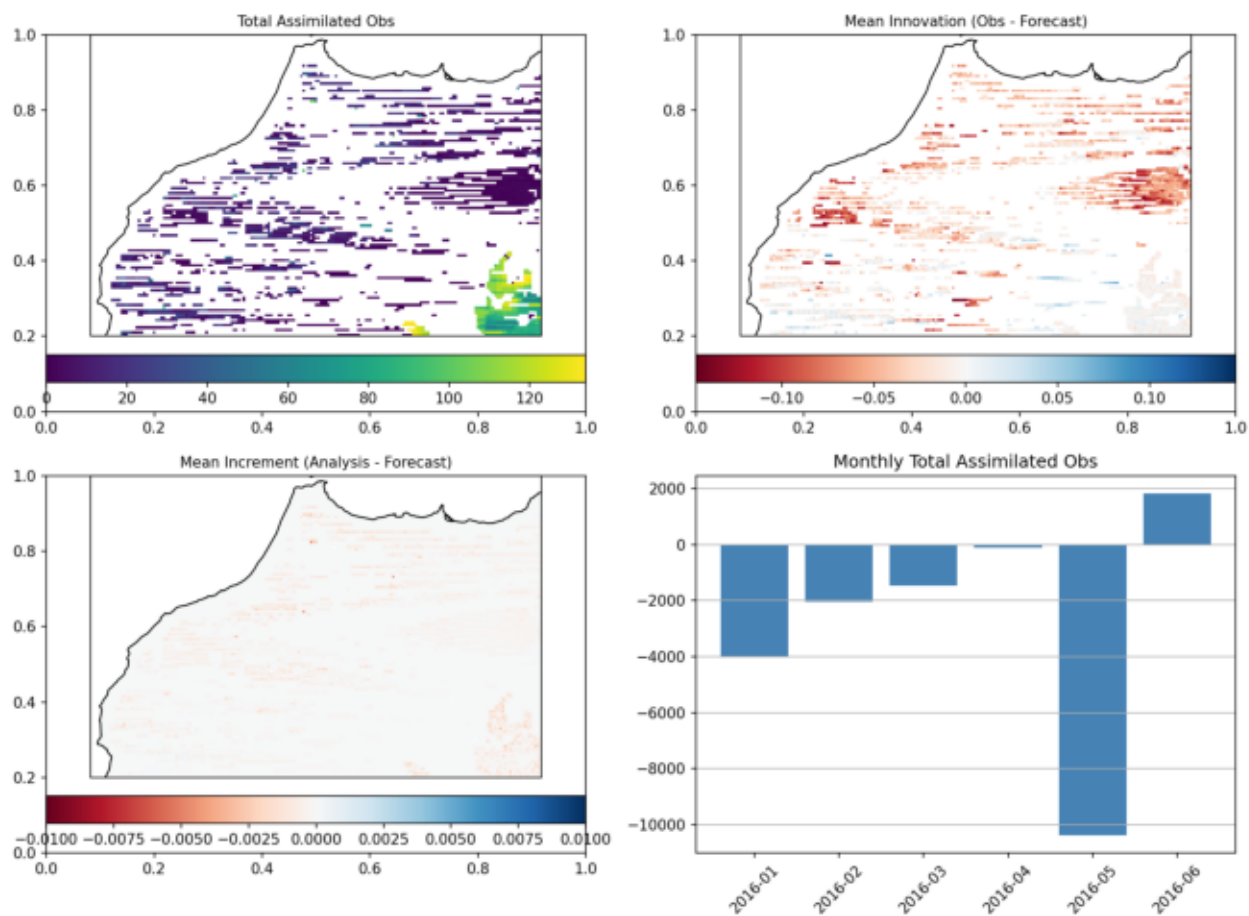


Figure 1: SMAP assimilation diagnostics showing spatial coverage, observation counts, and mean innovations/increments.

Figure 2 - Sensibilité CDF vs no-CDF (Jan-Jun 2016)

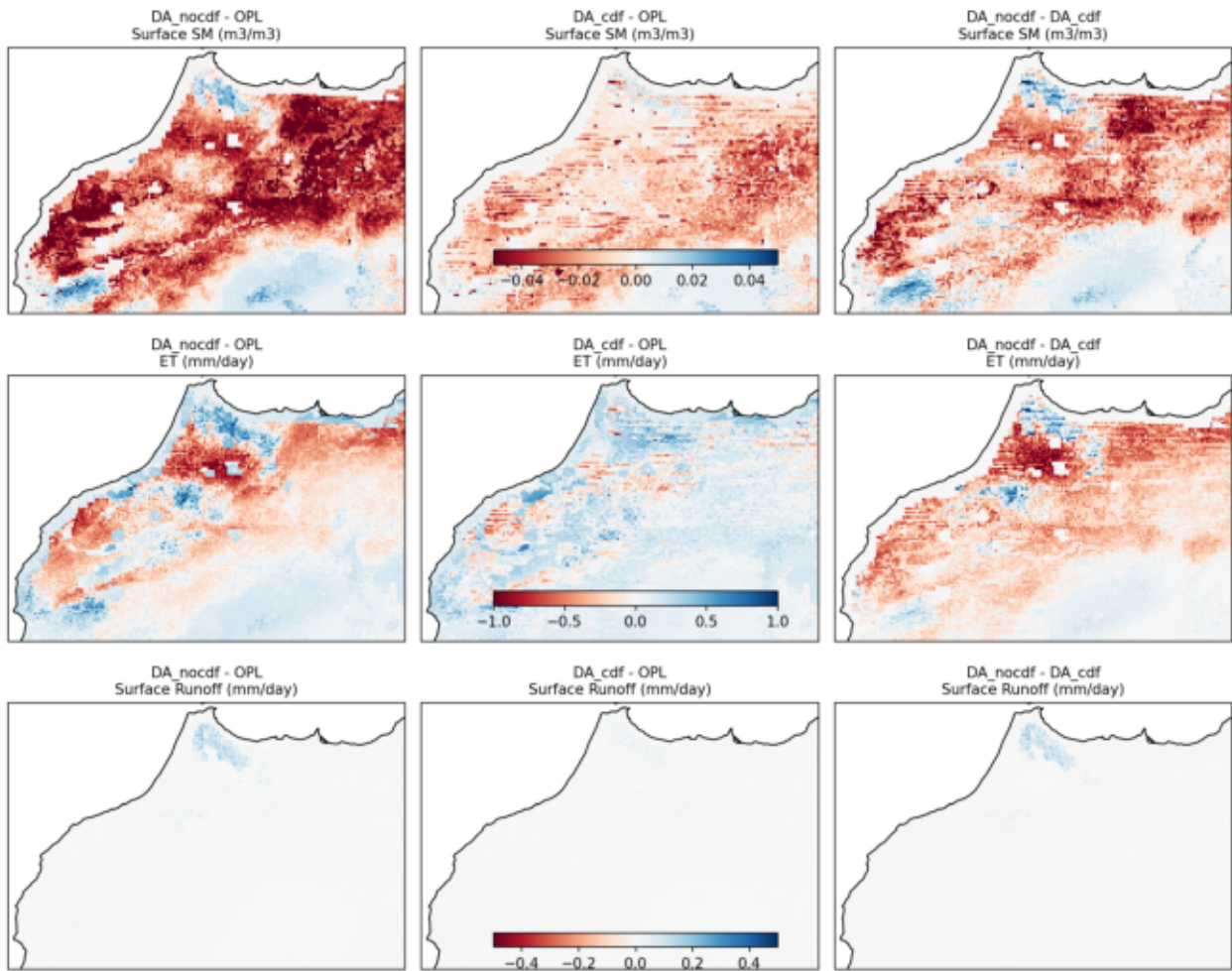


Figure 2: Sensitivity maps highlighting differences between DA (no-CDF), DA (CDF), and Open Loop.

Figure 3 - Vertical Propagation & Surface Flux Coherence (Jan-Jun 2016)

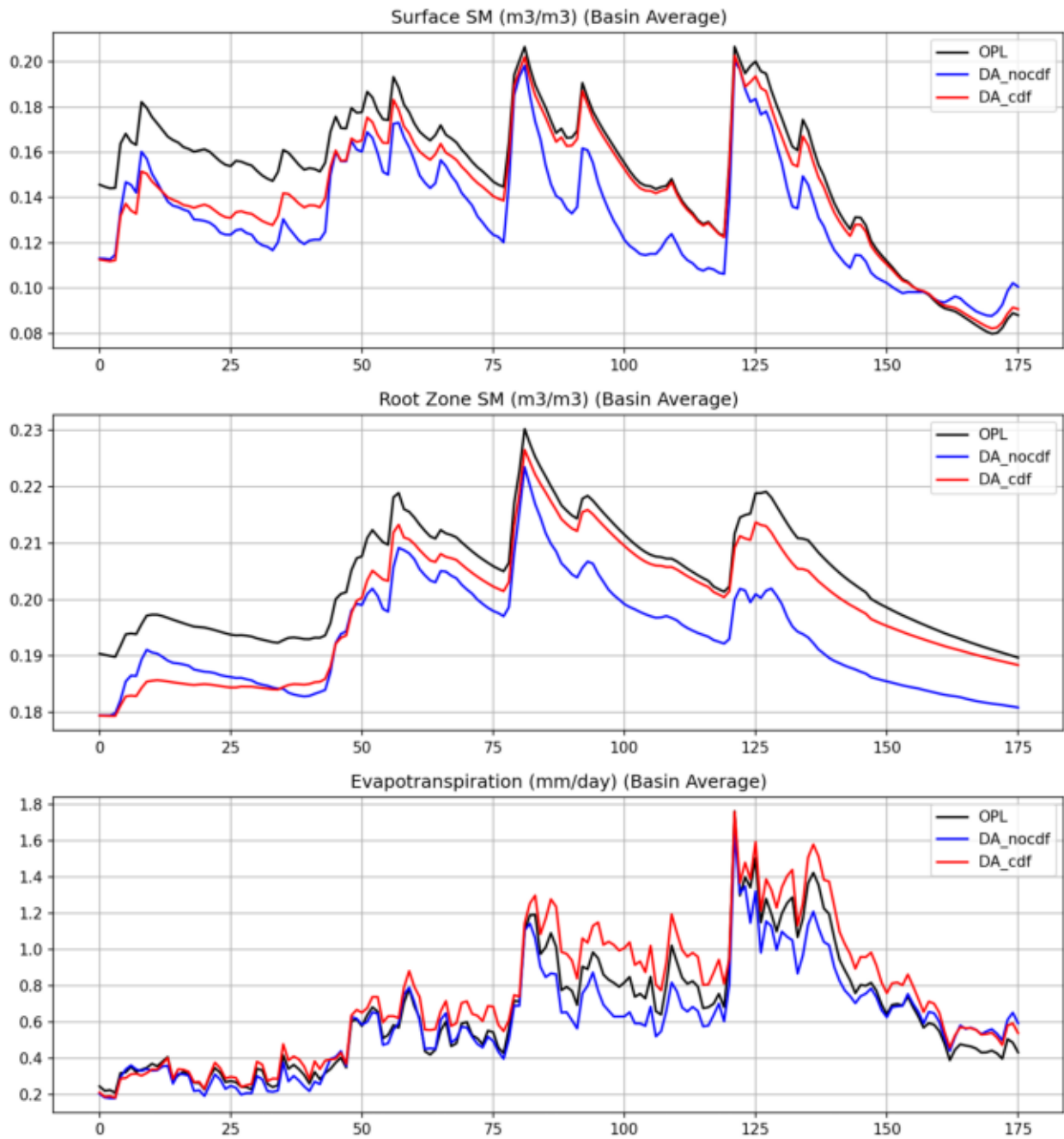


Figure 3: Vertical propagation of soil moisture updates into the root zone and coherence with ET fluxes.

Figure 4 - Runoff Partitioning (Jan-Jun 2016)

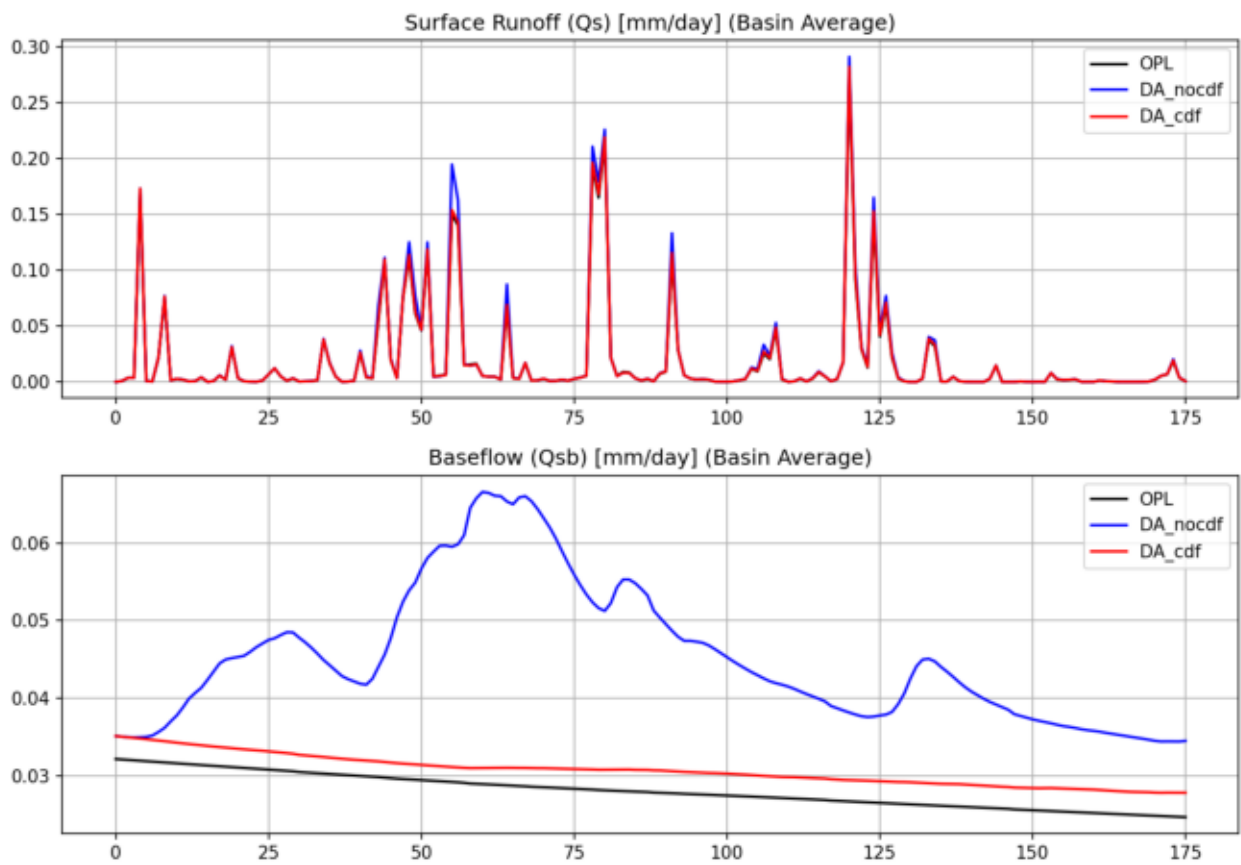


Figure 4: Impact of assimilation on runoff partitioning (surface vs baseflow) and seasonal dynamics.

Interpretation

Preliminary Interpretation:

- The assimilation of SMAP introduces specific spatial patterns of wetting/drying.
 - Vertical propagation from surface to root-zone heavily influences ET and runoff partitioning.
- The CDF matching process alters the absolute magnitudes of the increments compared to the no-CDF approach.

Diagnostics Status

Missing Diagnostics / Warnings:

No warnings. All expected data was found.